

UNIVERSITY OF THE ARMED FORCES - ESPE

Department of Exact Sciences

Object - Oriented Programming

NRC: 368663

HW09- ChickenFarmPersistence

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Homework 09 – Unit 1

Topic: Chicken Data Persistence Using JSON and CSV in Java, Python, and JavaScript

1. Java Implementation

1.1. ChickenFileManager Class

1.1.1. JSON Methods – saveJSON() and readJSON()

```java
public static void saveJSON(Chicken[] chickens) throws IOException {
    Gson gson = new Gson();

    FileWriter file = new FileWriter("chickens.json");
    gson.toJson(chickens, file);
    file.close();

    System.out.println("Chickens saved in JSON successfully!");
}
```

```java
public static Chicken[] readJSON() throws IOException {
    Gson gson = new Gson();

    FileReader file = new FileReader("chickens.json");
    Chicken[] chickens = gson.fromJson(file, Chicken[].class);
    file.close();

    return chickens;
}
```

1.1.2. CSV Methods – saveCSV() and readCSV()

```java
public static void saveCSV(Chicken[] chickens) throws IOException {
    FileWriter file = new FileWriter("chickens.csv");

    file.write("id,name,color,bornOnDate,ageInYears,molting\n");

    for (int i = 0; i < chickens.length; i++) {
        if (chickens[i] != null) {
            Chicken chicken = chickens[i];

            file.write(chicken.getId() + ","
                    + chicken.getName() + ","
                    + chicken.getColor() + ","
                    + chicken.getBornOnDate().getTime() + ","
                    + chicken.getAgeInYears() + ","
                    + chicken.isMolting() + "\n");
        }
    }

    file.close();

    System.out.println("Chickens saved in CSV successfully!");
}
```

```java
    public static Chicken[] readCSV() throws IOException {
        Chicken[] chickens = new Chicken[10];

        FileReader file = new FileReader("chickens.csv");
        Scanner scanner = new Scanner(file);

        scanner.nextLine();

        int i = 0;

        while (scanner.hasNextLine() && i < chickens.length) {
            String line = scanner.nextLine();

            if (line.isEmpty()) {
                continue;
            }

            String[] data = line.split(",");

            int id = Integer.parseInt(data[0]);
            String name = data[1];
            String color = data[2];
            Date bornOnDate = new Date(Long.parseLong(data[3]));
            int age = Integer.parseInt(data[4]);
            boolean molting = Boolean.parseBoolean(data[5]);

            chickens[i] = new Chicken(id, name, color, bornOnDate, molting);

            chickens[i].setAgeInYears(age);

            i++;
        }

        scanner.close();

        return chickens;
    }
}
```

1.2. ChickenFarmApp Class

1.2.1. Switch Menu Implementation

```java
// Menu
Scanner scanner = new Scanner(System.in);
int option;

do {
    System.out.println("\n===== CHICKEN FARM APP =====");
    System.out.println("1. Save Chickens");
    System.out.println("2. Read Chickens");
    System.out.println("3. Exit");
    System.out.print("Select an option: ");

    option = scanner.nextInt();

    switch (option) {
        case 1:
            ChickenFileManager.saveJSON(chickens);
            ChickenFileManager.saveCSV(chickens);
            break;

        case 2:
            System.out.println("\n--- CHICKENS FROM JSON ---");

            Chicken[] jsonChickens = ChickenFileManager.readJSON();

            if (jsonChickens != null) {
                for (int i = 0; i < jsonChickens.length; i++) {
                    if (jsonChickens[i] != null) {
                        System.out.println(jsonChickens[i]);
                    }
                }
            }

            System.out.println("\n--- CHICKENS FROM CSV ---");

            Chicken[] csvChickens = ChickenFileManager.readCSV();

            for (int i = 0; i < csvChickens.length; i++) {
                if (csvChickens[i] != null) {
                    System.out.println(csvChickens[i]);
                }
            }
            break;

        case 3:
            System.out.println("Goodbye!");
            break;

        default:
```

1.3. Execution Results

1.3.1. Saving Chickens

```
===== CHICKEN FARM APP =====
1. Save Chickens
2. Read Chickens
3. Exit
Select an option: 1
Chickens saved in JSON successfully!
Chickens saved in CSV successfully!
```

1.3.2. Reading Chickens

```
===== CHICKEN FARM APP =====
1. Save Chickens
2. Read Chickens
3. Exit
Select an option: 2
```

1.4. Generated Files

1.4.1. JSON File – chickens.json

```
--- CHICKENS FROM JSON ---
Chicken{id=0, name=chicken 1, color=white, bornOnDate=Sat Apr 03 00:00:00 GMT-05:00 2010, ageInYears=16, molting=true}
Chicken{id=1, name=chicken 2, color=white, bornOnDate=Sun Apr 03 00:00:00 GMT-05:00 2011, ageInYears=15, molting=true}
Chicken{id=2, name=chicken 3, color=white, bornOnDate=Tue Apr 03 00:00:00 GMT-05:00 2012, ageInYears=14, molting=true}
Chicken{id=3, name=chicken 4, color=white, bornOnDate=Wed Apr 03 00:00:00 GMT-05:00 2013, ageInYears=13, molting=true}
Chicken{id=4, name=chicken 5, color=white, bornOnDate=Thu Apr 03 00:00:00 GMT-05:00 2014, ageInYears=12, molting=true}
Chicken{id=5, name=chicken 6, color=white, bornOnDate=Fri Apr 03 00:00:00 GMT-05:00 2015, ageInYears=11, molting=true}
Chicken{id=6, name=chicken 7, color=white, bornOnDate=Sun Apr 03 00:00:00 GMT-05:00 2016, ageInYears=10, molting=true}
Chicken{id=7, name=chicken 8, color=white, bornOnDate=Mon Apr 03 00:00:00 GMT-05:00 2017, ageInYears=9, molting=true}
Chicken{id=8, name=chicken 9, color=white, bornOnDate=Tue Apr 03 00:00:00 GMT-05:00 2018, ageInYears=8, molting=true}
Chicken{id=9, name=chicken 10, color=white, bornOnDate=Wed Apr 03 00:00:00 GMT-05:00 2019, ageInYears=7, molting=true}
```

1.4.2. CSV File – chickens.csv

```
--- CHICKENS FROM CSV ---
Chicken{id=0, name=chicken 1, color=white, bornOnDate=Sat Apr 03 00:00:00 GMT-05:00 2010, ageInYears=16, molting=true}
Chicken{id=1, name=chicken 2, color=white, bornOnDate=Sun Apr 03 00:00:00 GMT-05:00 2011, ageInYears=15, molting=true}
Chicken{id=2, name=chicken 3, color=white, bornOnDate=Tue Apr 03 00:00:00 GMT-05:00 2012, ageInYears=14, molting=true}
Chicken{id=3, name=chicken 4, color=white, bornOnDate=Wed Apr 03 00:00:00 GMT-05:00 2013, ageInYears=13, molting=true}
Chicken{id=4, name=chicken 5, color=white, bornOnDate=Thu Apr 03 00:00:00 GMT-05:00 2014, ageInYears=12, molting=true}
Chicken{id=5, name=chicken 6, color=white, bornOnDate=Fri Apr 03 00:00:00 GMT-05:00 2015, ageInYears=11, molting=true}
Chicken{id=6, name=chicken 7, color=white, bornOnDate=Sun Apr 03 00:00:00 GMT-05:00 2016, ageInYears=10, molting=true}
Chicken{id=7, name=chicken 8, color=white, bornOnDate=Mon Apr 03 00:00:00 GMT-05:00 2017, ageInYears=9, molting=true}
Chicken{id=8, name=chicken 9, color=white, bornOnDate=Tue Apr 03 00:00:00 GMT-05:00 2018, ageInYears=8, molting=true}
Chicken{id=9, name=chicken 10, color=white, bornOnDate=Wed Apr 03 00:00:00 GMT-05:00 2019, ageInYears=7, molting=true}
```

---

2. Python Implementation

2.1. Chicken Class

2.2. File Management

2.2.1. JSON Save and Read Methods

```python
@staticmethod
    def saveJSON(chickens):
        data = []

        for chicken in chickens:
            if chicken is not None:
                data += [{
                    "id": chicken.getId(),
                    "name": chicken.getName(),
                    "color": chicken.getColor(),
                    "bornOnDate": chicken.getBornOnDate().isoformat(),
                    "ageInYears": chicken.getAgeInYears(),
                    "molting": chicken.isMolting()
                }]

        file = open("chickens.json", "w")
        json.dump(data, file, indent=4)
        file.close()

        print("Chickens saved in JSON successfully!")

    @staticmethod
    def readJSON():
        file = open("chickens.json", "r")
        data = json.load(file)
        file.close()

        chickens = [None] * len(data)

        for i in range(len(data)):
            item = data[i]

            bornOnDate = datetime.fromisoformat(item["bornOnDate"])

            chickens[i] = Chicken(
                item["id"],
                item["name"],
                item["color"],
                bornOnDate,
                item["molting"]
            )

            chickens[i].setAgeInYears(item["ageInYears"])

        return chickens
```

2.2.2. CSV Save and Read Methods

```python
@staticmethod
def saveCSV(chickens):
    file = open("chickens.csv", "w", newline="")

    writer = csv.writer(file)

    writer.writerow([
        "id", "name", "color",
        "bornOnDate", "ageInYears", "molting"
    ])

    for chicken in chickens:
        if chicken is not None:
            writer.writerow([
                chicken.getId(),
                chicken.getName(),
                chicken.getColor(),
                chicken.getBornOnDate().isoformat(),
                chicken.getAgeInYears(),
                chicken.isMolting()
            ])

    file.close()

    print("Chickens saved in CSV successfully!")


@staticmethod
def readCSV():
    chickens = [None] * 10

    file = open("chickens.csv", "r", newline="")
    reader = csv.reader(file)

    next(reader)

    i = 0

    for data in reader:
        if i < len(chickens):
            bornOnDate = datetime.fromisoformat(data[3])

            chickens[i] = Chicken(
                int(data[0]),
                data[1],
                data[2],
                bornOnDate,
                data[5].lower() == "true"
            )

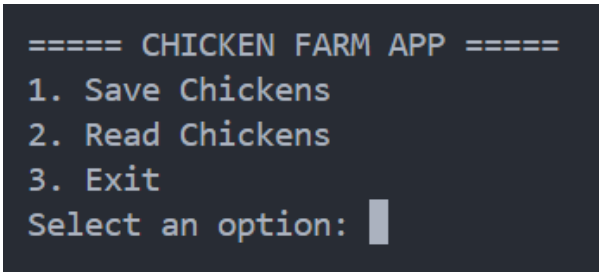
            chickens[i].setAgeInYears(int(data[4]))

            i += 1

    file.close()

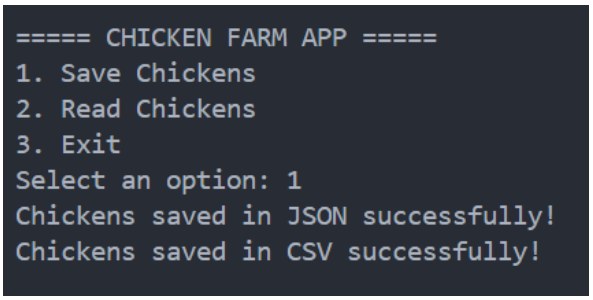
    return chickens
```

2.3. Main Program

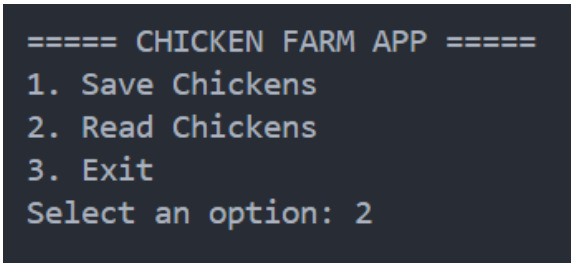


2.4. Execution Results

2.4.1. Saving Chickens



2.4.2. Reading Chickens



2.5. Generated Files

2.5.1. JSON File

```
--- CHICKENS FROM JSON ---
Chicken{id=0, name=chicken 1, color=white, bornOnDate=2010-04-03 00:00:00, ageInYears=16, molting=true}
Chicken{id=1, name=chicken 2, color=white, bornOnDate=2011-04-03 00:00:00, ageInYears=15, molting=true}
Chicken{id=2, name=chicken 3, color=white, bornOnDate=2012-04-03 00:00:00, ageInYears=14, molting=true}
Chicken{id=3, name=chicken 4, color=white, bornOnDate=2013-04-03 00:00:00, ageInYears=13, molting=true}
Chicken{id=4, name=chicken 5, color=white, bornOnDate=2014-04-03 00:00:00, ageInYears=12, molting=true}
Chicken{id=5, name=chicken 6, color=white, bornOnDate=2015-04-03 00:00:00, ageInYears=11, molting=true}
Chicken{id=6, name=chicken 7, color=white, bornOnDate=2016-04-03 00:00:00, ageInYears=10, molting=true}
Chicken{id=7, name=chicken 8, color=white, bornOnDate=2017-04-03 00:00:00, ageInYears=9, molting=true}
Chicken{id=8, name=chicken 9, color=white, bornOnDate=2018-04-03 00:00:00, ageInYears=8, molting=true}
Chicken{id=9, name=chicken 10, color=white, bornOnDate=2019-04-03 00:00:00, ageInYears=7, molting=true}
```

2.5.2. CSV File

```
--- CHICKENS FROM CSV ---
Chicken{id=0, name=chicken 1, color=white, bornOnDate=2010-04-03 00:00:00, ageInYears=16, molting=true}
Chicken{id=1, name=chicken 2, color=white, bornOnDate=2011-04-03 00:00:00, ageInYears=15, molting=true}
Chicken{id=2, name=chicken 3, color=white, bornOnDate=2012-04-03 00:00:00, ageInYears=14, molting=true}
Chicken{id=3, name=chicken 4, color=white, bornOnDate=2013-04-03 00:00:00, ageInYears=13, molting=true}
Chicken{id=4, name=chicken 5, color=white, bornOnDate=2014-04-03 00:00:00, ageInYears=12, molting=true}
Chicken{id=5, name=chicken 6, color=white, bornOnDate=2015-04-03 00:00:00, ageInYears=11, molting=true}
Chicken{id=6, name=chicken 7, color=white, bornOnDate=2016-04-03 00:00:00, ageInYears=10, molting=true}
Chicken{id=7, name=chicken 8, color=white, bornOnDate=2017-04-03 00:00:00, ageInYears=9, molting=true}
Chicken{id=8, name=chicken 9, color=white, bornOnDate=2018-04-03 00:00:00, ageInYears=8, molting=true}
Chicken{id=9, name=chicken 10, color=white, bornOnDate=2019-04-03 00:00:00, ageInYears=7, molting=true}
```

---

3. JavaScript Implementation

3.1. Chicken Class

3.2. File Management

3.2.1. JSON Save and Read Functions

```
static saveJSON(chickens) {
    let data = [];

    for (let i = 0; i < chickens.length; i++) {
      if (chickens[i] != null) {
        data[data.length] = chickens[i];
      }
    }

    let json = JSON.stringify(data, null, 4);

    fs.writeFileSync("chickens.json", json);

    console.log("Chickens saved in JSON successfully!");
  }

  static readJSON() {
    let json = fs.readFileSync("chickens.json", "utf8");
    let data = JSON.parse(json);

    let chickens = new Array(data.length).fill(null);

    for (let i = 0; i < data.length; i++) {
      let item = data[i];

      chickens[i] = new Chicken(
        item.id,
        item.name,
        item.color,
        new Date(item.bornOnDate),
        item.molting,
      );

      chickens[i].setAgeInYears(item.ageInYears);
    }

    return chickens;
  }
```

3.2.2. CSV Save and Read Functions

```
static saveCSV(chickens) {
    let csv = "id,name,color,bornOnDate,ageInYears,molting\n";

    for (let i = 0; i < chickens.length; i++) {
      if (chickens[i] != null) {
        let chicken = chickens[i];

        csv +=
          chicken.getId() +
          "," +
          chicken.getName() +
          "," +
          chicken.getColor() +
          "," +
          chicken.getBornOnDate().getTime() +
          "," +
          chicken.getAgeInYears() +
          "," +
          chicken.isMolting() +
          "\n";
      }
    }

    fs.writeFileSync("chickens.csv", csv);

    console.log("Chickens saved in CSV successfully!");
  }


  static readCSV() {
    let chickens = new Array(10).fill(null);

    let csv = fs.readFileSync("chickens.csv", "utf8");
    let lines = csv.trim().split("\n");

    let index = 0;

    for (let i = 1; i < lines.length && index < chickens.length; i++) {
      let data = lines[i].trim().split(",");

      let id = parseInt(data[0]);
      let name = data[1];
      let color = data[2];
      let bornOnDate = new Date(Number(data[3]));
      let age = parseInt(data[4]);
      let molting = data[5] === "true";

      chickens[index] = new Chicken(id, name, color, bornOnDate, molting);

      chickens[index].setAgeInYears(age);

      index++;
    }

    return chickens;
  }
}

module.exports = ChickenFileManager;

```

3.3. Main Program

```
// Menu
  const scanner = readline.createInterface({
    input: process.stdin,
    output: process.stdout,
  });

  function showMenu() {
    console.log("\n===== CHICKEN FARM APP =====");
    console.log("1. Save Chickens");
    console.log("2. Read Chickens");
    console.log("3. Exit");

    scanner.question("Select an option: ", function (answer) {
      let option = parseInt(answer);

      switch (option) {
        case 1:
          ChickenFileManager.saveJSON(chickens);
          ChickenFileManager.saveCSV(chickens);
          showMenu();
          break;

        case 2:
          console.log("\n--- CHICKENS FROM JSON ---");

          let jsonChickens = ChickenFileManager.readJSON();

          if (jsonChickens != null) {
            for (let i = 0; i < jsonChickens.length; i++) {
              if (jsonChickens[i] != null) {
                console.log(jsonChickens[i].toString());
              }
            }
          }

          console.log("\n--- CHICKENS FROM CSV ---");

          let csvChickens = ChickenFileManager.readCSV();

          for (let i = 0; i < csvChickens.length; i++) {
            if (csvChickens[i] != null) {
              console.log(csvChickens[i].toString());
            }
          }

          showMenu();
          break;

        case 3:
          console.log("Goodbye!");
          scanner.close();
          break;

        default:
          console.log("Invalid option");
          showMenu();
          break;
      }
    });
  }
```

3.4. Execution Results

3.4.1. Saving Chickens

```
===== CHICKEN FARM APP =====
1. Save Chickens
2. Read Chickens
3. Exit
Select an option: 1
Chickens saved in JSON successfully!
Chickens saved in CSV successfully!
```

3.4.2. Reading Chickens

```
===== CHICKEN FARM APP =====
1. Save Chickens
2. Read Chickens
3. Exit
Select an option: 2
```

3.5. Generated Files

3.5.1. JSON File

```
--- CHICKENS FROM JSON ---
Chicken{id=0, name=chicken 1, color=white, bornOnDate=Sat Apr 03 2010 00:00:00 GMT-0500 (hora estándar de Colombia), ageInYears=16, molting=true}
Chicken{id=1, name=chicken 2, color=white, bornOnDate=Sun Apr 03 2011 00:00:00 GMT-0500 (hora estándar de Colombia), ageInYears=15, molting=true}
Chicken{id=2, name=chicken 3, color=white, bornOnDate=Tue Apr 03 2012 00:00:00 GMT-0500 (hora estándar de Colombia), ageInYears=14, molting=true}
Chicken{id=3, name=chicken 4, color=white, bornOnDate=Wed Apr 03 2013 00:00:00 GMT-0500 (hora estándar de Colombia), ageInYears=13, molting=true}
Chicken{id=4, name=chicken 5, color=white, bornOnDate=Thu Apr 03 2014 00:00:00 GMT-0500 (hora estándar de Colombia), ageInYears=12, molting=true}
Chicken{id=5, name=chicken 6, color=white, bornOnDate=Fri Apr 03 2015 00:00:00 GMT-0500 (hora estándar de Colombia), ageInYears=11, molting=true}
Chicken{id=6, name=chicken 7, color=white, bornOnDate=Sun Apr 03 2016 00:00:00 GMT-0500 (hora estándar de Colombia), ageInYears=10, molting=true}
Chicken{id=7, name=chicken 8, color=white, bornOnDate=Mon Apr 03 2017 00:00:00 GMT-0500 (hora estándar de Colombia), ageInYears=9, molting=true}
Chicken{id=8, name=chicken 9, color=white, bornOnDate=Tue Apr 03 2018 00:00:00 GMT-0500 (hora estándar de Colombia), ageInYears=8, molting=true}
Chicken{id=9, name=chicken 10, color=white, bornOnDate=Wed Apr 03 2019 00:00:00 GMT-0500 (hora estándar de Colombia), ageInYears=7, molting=true}
```

3.5.2. CSV File

```
--- CHICKENS FROM CSV ---
Chicken{id=0, name=chicken 1, color=white, bornOnDate=Sat Apr 03 2010 00:00:00 GMT-0500 (hora estándar de Colombia), ageInYears=16, molting=true}
Chicken{id=1, name=chicken 2, color=white, bornOnDate=Sun Apr 03 2011 00:00:00 GMT-0500 (hora estándar de Colombia), ageInYears=15, molting=true}
Chicken{id=2, name=chicken 3, color=white, bornOnDate=Tue Apr 03 2012 00:00:00 GMT-0500 (hora estándar de Colombia), ageInYears=14, molting=true}
Chicken{id=3, name=chicken 4, color=white, bornOnDate=Wed Apr 03 2013 00:00:00 GMT-0500 (hora estándar de Colombia), ageInYears=13, molting=true}
Chicken{id=4, name=chicken 5, color=white, bornOnDate=Thu Apr 03 2014 00:00:00 GMT-0500 (hora estándar de Colombia), ageInYears=12, molting=true}
Chicken{id=5, name=chicken 6, color=white, bornOnDate=Fri Apr 03 2015 00:00:00 GMT-0500 (hora estándar de Colombia), ageInYears=11, molting=true}
Chicken{id=6, name=chicken 7, color=white, bornOnDate=Sun Apr 03 2016 00:00:00 GMT-0500 (hora estándar de Colombia), ageInYears=10, molting=true}
Chicken{id=7, name=chicken 8, color=white, bornOnDate=Mon Apr 03 2017 00:00:00 GMT-0500 (hora estándar de Colombia), ageInYears=9, molting=true}
Chicken{id=8, name=chicken 9, color=white, bornOnDate=Tue Apr 03 2018 00:00:00 GMT-0500 (hora estándar de Colombia), ageInYears=8, molting=true}
Chicken{id=9, name=chicken 10, color=white, bornOnDate=Wed Apr 03 2019 00:00:00 GMT-0500 (hora estándar de Colombia), ageInYears=7, molting=true}
```
